Proceedings of the 18th World Congress
The International Federation of Automatic Control
Milano (Italy) August 28 - September 2, 2011

Modeling of the aromatic profile in wine-making fermentation: the backbone equations

R. David [*] **D. Dochain** [*,⋆] **J.-R. Mouret** [**]
A. Vande Wouwer [***] **J.-M. Sablayrolles** [**]

[*] *CESAME, UCL, 4 Avenue Georges Lemaître, Louvain-la-Neuve, Belgium (e-mail: {robert.david, denis.dochain}@uclouvain.be).*
[**] *UMR 1083 SPO, INRA SUPAGRO, 2 Place Viala, 34060 Montpellier cedex, France (e-mail: {mouretj, sablayro}@supagro.inra.fr)*
[***] *Automatic Control Laboratory, UMONS, 31 Boulevard Dolez, Mons, Belgium (e-mail: alain.vandewouwer@umons.ac.be)*

Abstract: The aromatic profile of a wine is mainly determined during the grape-must fermentation and is characterized by several compounds called flavour markers. These particular compounds are minority by-products produced from "leaks of metabolism" of the used yeast. The final objective of this work is to gain more insight about the synthesis of the aromatic profile in order to optimize it. For this purpose, a first necessary step is the development of a model representing the main physiological phenomena observed during the batch fermentation in the wine-making process in order to later extend it with flavour-markers equations. The main-kinetics model described in this paper is based on a set of biological reactions in which nitrogen compounds such as hexose transporters play a central role, in line with experimental evidence deduced from extensive experimental studies (Malherbe, 2003).

Keywords: Wine-making, fermentation, batch reactor, dynamical model, nitrogen, hexose transporters, yeast activity

1. INTRODUCTION

Understanding the underlying kinetic mechanisms of bioprocesses is required to develop mathematical models and simulation tools providing efficient control algorithms so as to maximize the process quality (Bastin and Dochain, 1990). Following this line of thought, the development of paradigms in food process control is one of the issues of the EC CAFÉ project (`www.cafe-project.org`) that considers four case studies, and among them the bioconversion process of wine-making mainly described by the alcoholic fermentation step.

The objective of the CAFÉ project about the wine-making fermentation is to design control tools aimed at optimising the fermentation so as to obtain a well defined aromatic profile. Indeed, during the alcoholic fermentation hexoses (glucose and fructose) are converted to ethanol and carbon dioxide, but many other compounds are removed from the must and a large set of by-products are formed that affect the organoleptic properties of the wine (Swiegers et al., 2005). These by-products (higher alcohols, esters, sulphur compounds) represent less than 4 % of the yeast production whereas the ethanol and the glycerol represent respectively more than 90 % and around 5 % of the production.

Before considering these minority compounds a relevant description of the main kinetics is needed as they lead the fermentation behaviour. Different approaches can be considered to provide reliable information about the ethanolic fermentation process dynamics. A first a priori classical modelling approach is to consider mass balances of the key components combined with an appropriate knowledge about the process operation and dynamics. Another a priori attractive approach may be to consider microbiological knowledge about the production of the important organoleptic compounds of the wine in order to develop models that might link the metabolic pathways knowledge to the process operation. If the second approach remains a challenging task (e.g. Charnomordic et al. (2010)), the first approach has been the object of extensive studies over the past decades. The literature review and experimental observations have led us to the conclusion that

- the first step in the fermentation is the catalyst synthesis, namely the growth of yeast with the nitrogen amount as limiting substrate;
- the second one is the catalysis, namely the degradation of the non-limiting sugar substrate into ethanol and carbon dioxide.

This therefore enlightened the modeling scheme presented in the next sections.

As mentioned in a previous paper (David et al., 2010), the first comprehensive kinetic models (Boulton, 1980;

[⋆] Honorary Research Director FNRS, Belgium

Sevely et al., 1981; Williams et al., 1986; Caro et al., 1991; Remedios Marín, 1999) were describing the influence of sugar and ethanol levels, and of temperature on sugar utilization, capturing the general macroscopic trends found in practice. Models have been developed to predict the transition from normal to sluggish or stuck fermentation [1] , with kinetics based on nitrogen as a growth-limiting nutrient, in isothermal conditions (Cramer et al., 2002), and later by including the temperature dependency of some parameters (Coleman et al., 2007). Unfortunately, the relevance of these models has never been validated in real wine-making conditions.

Several more empirical or non-parametric models have also been published and the model of Malherbe et al. (2004), largely based on the first principle modeling approach, considers the main yeast physiological mechanisms:

(1) glucose transport, inhibited by ethanol $E(t)$ and the limiting feature of fermentation;
(2) glycolysis, *i.e.* glucose degradation into ethanol and CO_2, which is not limiting;
(3) nitrogen transport, also strongly inhibited by ethanol;
(4) synthesis of glucose transporters from a fraction of the absorbed nitrogen.

The model predicts the rate at which glucose is consumed and the amount of ethanol or CO_2 produced. It includes the effects of the main involved factors : temperature $T(t)$, which can vary within a predefined range (18 to 30°C) and assimilable nitrogen $N(t)$, which has a major impact on the yeast activity and varies a lot according to the musts. This latter consideration is here primordial as nitrogen is the source of the flavour-compounds precursors. The model accurately describes the fermentation kinetics of more than 80% of a large number of experiments performed with 20 wine yeast strains, 69 musts and different fermentation conditions. Thanks to the wide domain of validity of the model, a simulator was developed to help winemakers to optimize tank management.

However this model does not include consistent mass balances. With that respect its extension to new compounds is difficult because the equations formulation is complex and the large number of parameters does not allow a straightforward identification (if a modification due to operational conditions is necessary). Nevertheless the resulting simulator can be used to produce an easy-to-use experimental database describing the main kinetics, given the significant results provided by this model and the involved knowledge of the process it required.

The next section presents the development of the model equations whereas Section 3 stresses the needed assumptions to include the variable called excess-transporters concentration. The parameters identification is described in the following (Section 4) and the model is validated in Section 5. Some conclusions are drawn in Section 6.

[1] Between one and two weeks are typically necessary to achieve a complete fermentation, but the process can take significantly longer to complete (sluggish fermentation) or can leave an important residual sugar level (stuck fermentation).

2. MODEL FORMULATION

A first step in the modeling of the fermentation process is to provide a relevant description of the main kinetics involving the growth of biomass on nitrogen and the production of ethanol and carbon dioxide resulting from the sugar consumption. This mathematical model can be extended afterwards to the description of the considered flavour-active compounds.

2.1 Preliminary

The concentrations of glucose $S(t)$ and ethanol $E(t)$, and their time variations $\dot{S}$ and $\dot{E}$ can be deduced from the released amount of carbon dioxide $CO_2(t)$ assuming Gay-Lussac-like relations (Malherbe et al., 2004):

$$S(t) = S(0) - 2.17\, CO_2(t), \text{ with } CO_2(0) = 0 \quad (1)$$

$$E(t) = 0.464\,(S(0) - S(t)), \text{ with } E(0) = 0 \quad (2)$$

leading to

$$E(t) = CO_2(t) \Longleftrightarrow \dot{E}(t) = \dot{CO_2}(t) \quad (3)$$

$$\dot{S}(t) = -2.17\,\dot{CO_2}(t) \quad (4)$$

The reaction scheme commonly used in fermentation considers the growth of biomass on nitrogen and sugar, and the synthesis of ethanol from sugar (Sonnleitner and Käppeli, 1986). Nevertheless, the sugar is not limiting in the present context. Therefore, the first reaction scheme that has been considered (David et al., 2010) is the following: biomass X grows on nitrogen N (the limiting nutrient in the fermentation process), and sugar S is enzymatically degraded into ethanol E and CO_2, and inhibited by ethanol, providing

$$k_1\, N \longrightarrow X \quad (5)$$

$$k_2\, S \xrightarrow{X} E + CO_2 \quad (6)$$

where the coefficients k_1 and k_2 represent the yield coefficients respectively associated to nitrogen and sugar consumptions. The value of k_2 can already be deduced from (4).

Experiments used to design the Malherbe model have underlined the influence of the initial concentration of nitrogen N_0 on the batch process. This latter have even been used as a parameter of the differential equations, which cannot be the case in a mass balance design of the process.

The influence of N_0 is illustrated in figure 1: it can be observed that the maximum value of biomass concentration X_{max} does not evolve linearly with N_0 in practice (solid curve), as it could be expected and as it is actually illustrated by our first mass balance model ($\cdot-$ curve). Indeed, the model formulation implies a complete consumption of the nitrogen for the biomass growth and consequently a linear relation between X_{max} and N_0, whatever the considered kinetic model: Michaëlis-Menten, Haldane, ... (only the transient behaviour will change). This observation leads us to the introduction of another reaction to characterize the nitrogen consumption into the model.

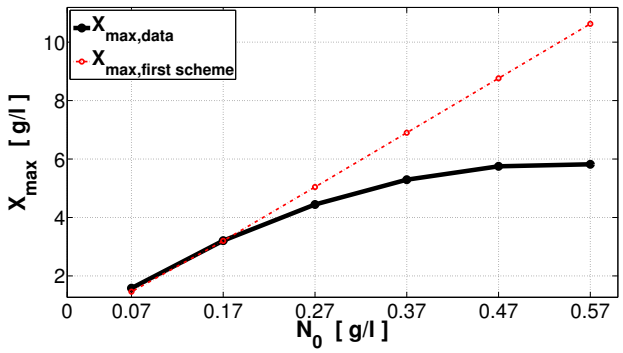


Fig. 1. Evolution of the maximum value of biomass X_{max} depending on the initial concentration of nitrogen N_0 with the Malherbe et al. (2004) simulator (black) and with the first mass balance model $(\cdot -)$ from David et al. (2010)

In wine-making conditions, a part of nitrogen is assimilated so as to synthesize new yeast cells but the remaining part is mostly used in the synthesis and repairing of essential proteins, see Figure 2. Indeed, the yeast cell can assimilate glucose and nitrogen thanks to dedicated proteins called transporters which allow the membrane crossing.

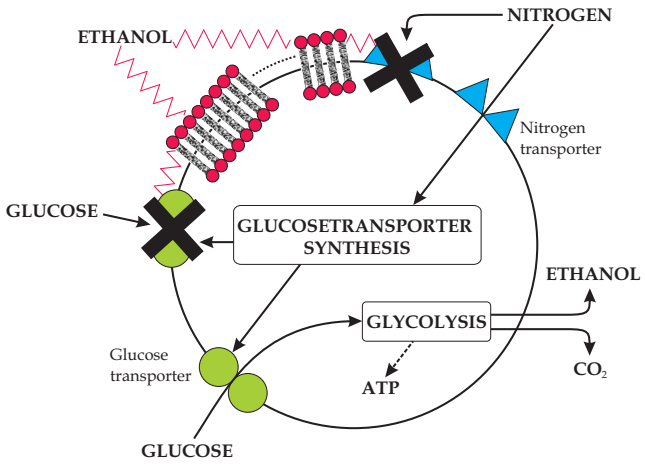


Fig. 2. Yeast activity scheme

During the fermentation these transport proteins have to sustain a catabolic inactivation by the ethanol. This phenomenon increases with the ethanol concentration and impacts the sugar transport, hence the fermentation kinetic is slacked (decrease of CO_2 production rate). Experiments have shown that when the initial nitrogen concentration is low, the yeast mainly focus on the cells production. When N_0 is larger, the cells production increases but at the same time more nitrogen is used for the transporters synthesis so as to prevent their catabolic inactivation (Malherbe, 2003). This effect has also been experimentally illustrated by adding assimilable nitrogen in the fermentor once the biomass is at steady-state: the regrowth is weak but the yeast activity is reboosted (peak of CO_2 production rate $\dot{CO}_2(t)$).

2.2 Model equations

The current model of the main kinetics is based on the following scheme of biochemical conversion, and build upon the work of Malherbe et al. (2004) and David et al. (2010):

$$k_1\, N_x \longrightarrow X \tag{7}$$

$$k_1'\, N_{tr} \xrightarrow{X} Tr \tag{8}$$

$$k_2\, S \xrightarrow{X,\,Tr} E + CO_2 \tag{9}$$

where the biomass X (yeast) grows on nitrogen dedicated to the biomass growth N_x. Tr represents the concentration of glucose transporters in excess, as it is explained in the assumptions (Section 3). N_{tr} is the nitrogen dedicated to the excess-transporters synthesis. Sugar S is enzymatically degraded, by the yeast and with the help of transporters, into ethanol E and carbon dioxide CO_2.

The above reaction scheme can be described mathematically in the following set of mass balances:

$$\dot{X} = \mu_{max}(T)\,\frac{N_x}{K_x\,X + N_x}\,X \tag{10}$$

$$\dot{N}_x = -k_1\,\dot{X} \tag{11}$$

$$\dot{Tr} = \eta_{max}(T)\,\frac{N_{tr}}{K_{tr} + N_{tr}}\,X \tag{12}$$

$$\dot{N}_{tr} = -k_1'\,\dot{Tr} \tag{13}$$

$$\dot{E} = \dot{CO}_2 = \beta_{max}(T)\,\frac{S}{K_S + S}\,\frac{K_E(T)}{K_E(T) + E}\,X\,(1 + \phi(T)\,Tr) \tag{14}$$

$$\dot{S} = -k_2\,\dot{CO}_2 \tag{15}$$

with

$$\begin{array}{ll} X_0 = X(t=0), & N_{x,0} = N_x(t=0), \\ Tr_0 = 0, & N_{tr,0} = N_{tr}(t=0), \\ E_0 = CO_{2,0} = 0, & S_0 = S(t=0) \end{array} \tag{16}$$

$\mu_{max}(T)$, $\eta_{max}(T)$ and $\beta_{max}(T)$ are the maximum specific reaction rates. K_x is the Contois constant, and K_{tr}, K_S are the Michaelis-Menten constants. $K_E(T)$ represents the ethanol inhibition. k_1, k_1', k_2 are stoichiometric coefficients. $\phi(T)$ is an efficiency parameter related to Tr and associated to the fermentation activity (ethanol or CO_2 production rate).

The initial conditions are of common values for wine-making conditions: $X_0 = 10^9$ cells/l, $N_0 = [0.07 : 0.57]$ g/l, $S_0 = 200$ g/l.

The assumptions described in Section 3 allow to determine $N_{x,0}$ and $N_{tr,0}$ thanks to N_0, and to generate 'experimental' data for the Tr parameters to be identified.

The temperature dependency is determined by identification of the concerned parameters with experiments made at different temperatures in the range [18:30]°C (see the table 1 in Section 4).

3. MODEL ASSUMPTIONS

There is no information into the literature about the dynamical behaviour of the proteins called glucose transporters,

neither about their concentration at steady-state (if any) in the wine-making case or other. Therefore we had to extract by ourselves some knowledge on these compounds, and assumptions were needed to define steady-state values and transient behaviours so as to generate 'data' to be predicted by the model (parameters identification of (12) and (13)).

3.1 Initial values of N_x and N_{tr}

The variable Tr represents an excess-transporters concentration determined by a reference experiment.

This latter one is defined by a very low initial concentration of nitrogen: 0.07 g/l but this allows a normal fermentation. It means that a minimum amount of transporters is synthesized so as to degrade all the sugar into ethanol and carbon dioxide.

For this particular experiment, the ratio N_0/X_{max} (X at steady state) is calculated, giving the proportion of nitrogen dedicated to biomass growth and synthesis of the minimum amount of transporters needed for a normal fermentation:

$$\alpha = \frac{0.07}{X_{max,0.07}}$$

As illustrated in figure 3 this ratio can be applied to the other X_{max} obtained with different N_0 and thus provide the vector $N_{x,0}$, i.e. the initial nitrogen concentration dedicated to biomass growth and synthesis of the minimum amount of transporters.

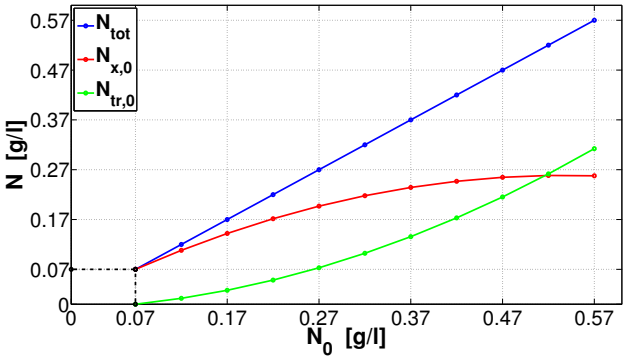


Fig. 3. Initial nitrogen distribution following their use

It can be mathematically summarized by:

$$N_{x,0} = \alpha \cdot X_{max}(N_0), \text{ for } \mathrm{N_0} = 0.07 \rightarrow 0.57\mathrm{g/l}$$

Therefore the initial nitrogen concentration dedicated to excess-transporters synthesis can be deduced by

$$N_{tr,0} = N_0 - N_{x,0}$$

as shown in figure 3. For $N_0 = 0.07$ g/l, $N_{tr,0}$ is naturally equal to 0.

3.2 Steady-state value Tr_{max}

It is of common practice to consider that into proteins there is 16% of nitrogen (Jones, 1941). This consideration can be directly applied in our case to determine the Tr_{max} (steady-state value) corresponding to $N_{tr,0}$. See figure 4. Mathematically,

$$Tr_{max} = \frac{N_{tr,0}(N_0)}{0.16}, \text{ for } \mathrm{N_0} = 0.07 \rightarrow 0.57\mathrm{g/l} \qquad (17)$$

As a consequence k_1' is determined and equal to 0.16.

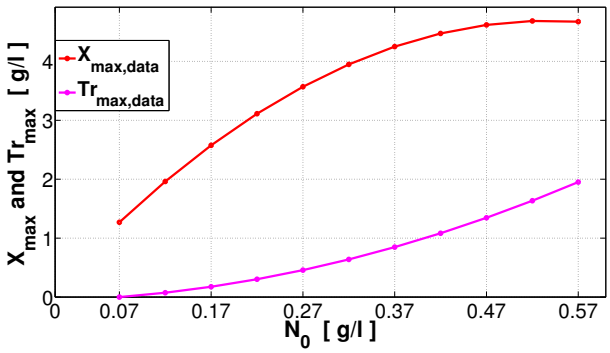


Fig. 4. Steady-state values of X and Tr related to the initial concentration of nitrogen N_0

3.3 Transient behaviour of Tr

As transporters are proteins synthesized by yeast, their transient behaviour is considered as the same as the biomass and is scaled to the corresponding Tr_{max} value.

4. MODEL-PARAMETERS IDENTIFICATION

Data generated by the Malherbe simulator and the assumptions of section 3 have been used to estimate the parameters of the equations (10-15). These parameters have been identified using the Nelder-Mead (simplex) method implemented in the Matlab function *fminsearch*, and the ODE solver *ode15s*. The model has a cascade structure which makes easier the identification task: the parameters $\mu_{max}(T)$, K_x and k_1 are identified, then $\eta_{max}(T)$, K_{tr}, and finally $\beta_{max}(T)$, K_S, K_E and ϕ.

The values of $\eta_{max}(T)$, K_{tr} and $\phi(T)$ can be just considered as of reasonable values face to the other parameters of the model as no source can provide information about the dynamical behaviour of the transporters.

Table 1. *Model-parameters values*

Parameter	Range of value	Unit
T	18 – 30	°C
$\mu_{max}(T)$	0.141 – 0.485	1/h
K_x	0.0558	-
k_1	0.0556	-
$\eta_{max}(T)$	0.061 – 0.212	1/h
K_{tr}	0.174	g/l
k_1'	0.160 (see section 3.2)	-
$\beta_{max}(T)$	0.795 – 1.027	g/l
K_S	3.699	g/l
$K_E(T)$	10.680 – 22.491	g/l
$\phi(T)$	0.261 – 0.698	-
k_2	2.170 (see section 2.1)	-

5. MODEL VALIDATION

The model of the main kinetics of fermentation has been validated with relevant data concerning X, N, S, E and CO_2. However the variable Tr has been entirely deduced from observations and assumptions, and cannot be supported by literature data.

Some results are stressed in figures 5 and 6 corresponding to extreme values in the considered ranges: [$T = 18$ °C; $N_0 = 0.07$ g/l] and [$T = 30$ °C; $N_0 = 0.57$ g/l].

It can be observed that Tr is equal to zero on figure 5 as $N_{tr,0} = 0$ g/l, and that the concentration of Tr is maximal for $N_0 = 0.57$ g/l.

The new simulator fits pretty well the 'data' originated by the simulator of Malherbe et al., knowing that it is not easy to fit a variable like a time derivative ($\dot{CO_2}$).

6. CONCLUSION

The wine-making fermentation involves complex biomechanisms and an efficient mass balance modeling of the key components requires a relevant experimental database. This experiment background has been synthesized in the logistic model of Malherbe et al. (2004) and the resulting simulator provides a representative behaviour of the process. Nevertheless this model is not suitable for the control purposes invoked in the CAFÉ project as it is not easy to manipulate and adapt (numerous parameters and initial conditions used as parameters). A mass balance model is therefore developed by using the easy-to-use database provided by the Malherbe et al. simulator.

The model suitably describes the fermentation for different temperatures and with different initial nitrogen concentrations. This results from physiological considerations such as the growth of biomass on nitrogen and the degradation of sugar into ethanol and carbon dioxide, but also by accounting the role of particuliar proteins called transporters. These transporters are involved in the second main reaction of nitrogen consumption and moreover impact the fermentation activity (represented by $\dot{CO_2}$).

Assumptions were needed to generate representative data of the excess-transporters concentration and influenced the model development, so as to finally obtain a mathematical model where the nitrogen is distributed following its use into biomass growth and excess-transporters synthesis.

A next step will be completed by validating the model on real experimental data and by extending it with a dynamical description of the flavour compounds. The effect of a nitrogen addition during the fermentation will also be hopefully integrated as a further development.

ACKNOWLEDGEMENTS

This paper includes results of the CAFÉ project that is supported by the Food, Agriculture and Fisheries, and Biotechnology program of the European Community (Contract number KBBE-212754). It also presents research results of the Belgian Programme on Interuniversity Poles of Attraction initiated by the Belgian State, Prime Minister's Office, Science, Technology and Culture. The scientific responsibility rests with its authors.

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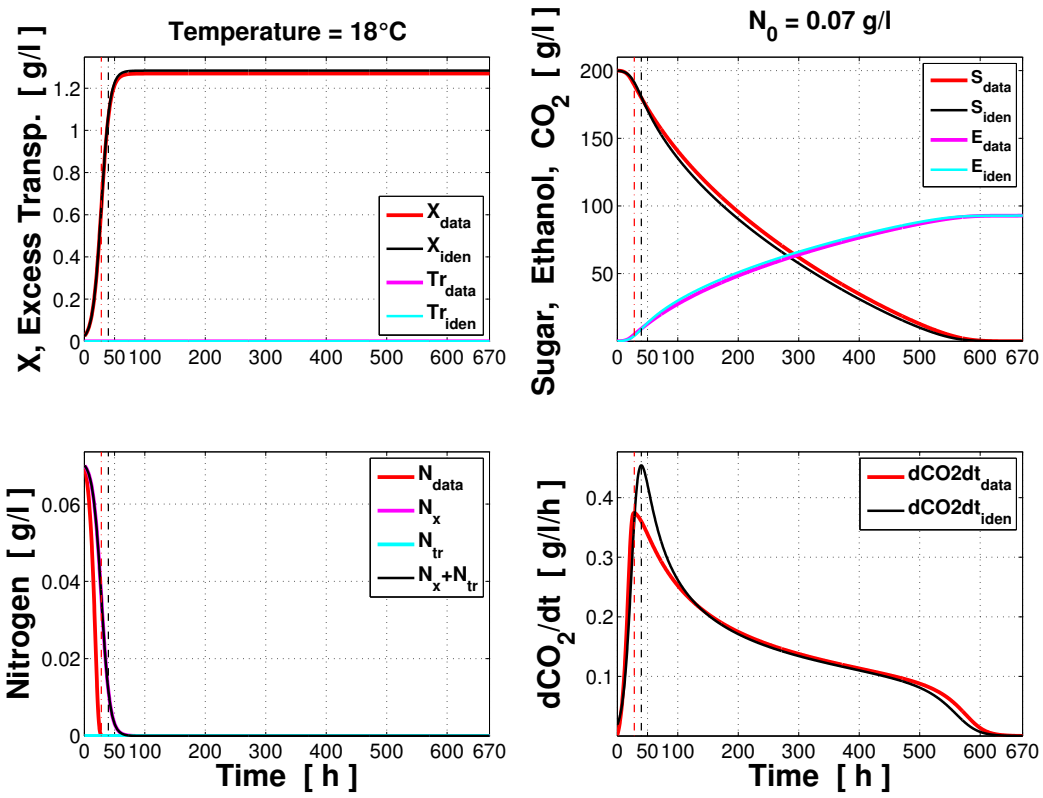


Fig. 5. Resulting curves of the new simulator ($*_{iden}$) face to the curves of the Malherbe *et al.* simulator ($*_{data}$) (Malherbe et al., 2004). The vertical lines correspond to the respective instants when $\dot{CO}_2 = \max(\dot{CO}_2)$, see last sub-figure

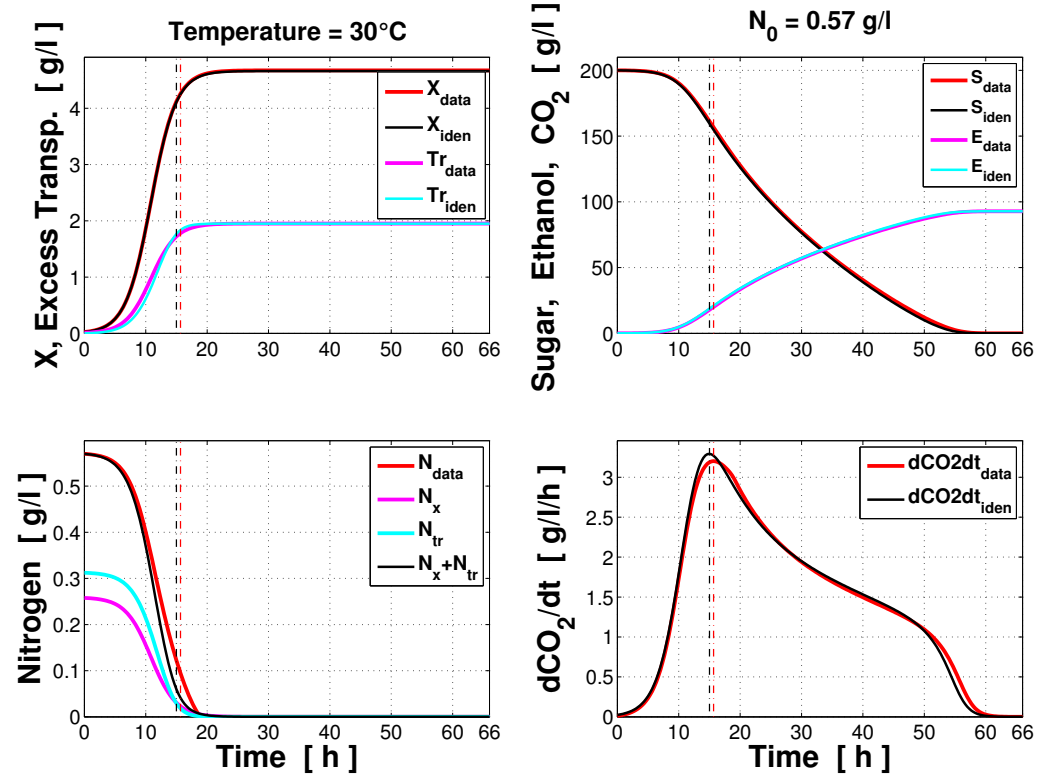


Fig. 6. Resulting curves of the new simulator ($*_{iden}$) face to the curves of the Malherbe *et al.* simulator ($*_{data}$) (Malherbe et al., 2004). The vertical lines correspond to the respective instants when $\dot{CO}_2 = \max(\dot{CO}_2)$, see last sub-figure